MILITARY SPECIFICATION

HEADSET-MICROPHONE, SOUND POWERED H-159/GIC-10

1. SCOPE

1.1 This specification covers one type of headset-microphone, designated Headset-Microphone, Sound Powered H-159/GIC-10, which is one item of Intercommunication Set, Sound Powered AN/GIC-10.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

J-C-98	Cable and Wire, Insulated; Methods of Sampling and Testing

Military

MIL-I-4997	Insulating and Jacketing Compounds for Use in Cords, Cordages, and Cables
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-T-9107	Test Reports, Preparation of
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-R-25717	Reliability Assurance Program for Electronic Equipment
MIL-I-27185	Intercommunication Set, Sound Powered AN/GIC-10
MIL-C-27187	Connector-Switch U-177/U

DRAWINGS

Air Force

58B12627	Simulator-Microphone Signal, Outline Dimensions of
59D12685	Intercommunication Set, Sound Powered AN/GIC-10 - Assembly of
59D12688	Headset-Microphone, Sound Powered H-159/GIC-10 - Assembly of

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PUBLICATIONS

USAF Specification Bulletin

329 Codes for Aircraft Finishing Schemes

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Standards Association

ASA Standard Z24.9-1949	American Standard Method for the Coupler Calibration of Earphones
ASA Standard Z24.22-1957	American Standard Method for the Measurement of Real Ear Attenuation of Ear Protectors at Threshold

(Application for copies should be addressed to the American Standards Association, 70 East 45th St., N. Y. 17, N. Y.)

Advisory Group on Reliability of Electronics Equipment (AGREE)
Office of Ass't. Secretary of Defense (Research and Engineering)
Report, 4 June 1957--Reliability of Military Electronic Equipment

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Qualification Testing.- The headset-microphone furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Components.- Headset-Microphone, Sound Powered H-159/GIC-10 shall consist of the following components assembled in accordance with Drawing No. 59D12688:

1 each	Headband, Headset
2 each	Earcup, Sound Protector
2 each	Earphone, Sound Powered H-181/GIC-10
1 each	Microphone, Sound Powered M-89/GIC-10
1 each	Cord Assembly, terminated with Connector-Switch U-177/U, in accordance with MIL-C-27187

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3.3 General Specification.- The requirements of MIL-E-5400 for class 1 equipment apply as requirements of this specification. When the two specifications conflict, this specification shall govern.

3.3.1 Approval of Nonstandard Parts and Material.- The requirements of MIL-E-5400 for nonstandard parts and materials shall not apply. Performance in accordance with the reliability tests specified herein shall be considered to constitute approval of nonstandard parts or materials.

3.3.2 Identification of Product.- The marking shall be applied by a suitable plastic sleeve. Marking shall include only the following information:

Headset-Microphone, Sound Powered H-159/GIC-10
Order No. ____*_____ (Government order number)
*Data to be inserted by the contractor

3.3 Interference.- Interference control requirements of MIL-E-5400 shall not apply.

3.3.4 Life.- The life requirements of MIL-E-5400 shall not apply.

3.4 Standard Sample.- When stated in the contract or purchase order, a model equipment, Headset-Microphone H-159/GIC-10, will be made available to the contractor on loan to be used as a general guide except in instances where specific improvements are detailed herein or where requirements dictate changes. The model equipment shall not be subjected to any destructive tests nor shall any part or component with the exception of subassemblies retained by machine screws, or the like, be removed or disassembled.

3.5 Reliability.- The reliability requirements of MIL-R-25717 shall apply as requirements of this specification with the following additions and exceptions:

3.5.1 Mean-Time-Between-Failure (MTBF) Requirements.- The H-159/GIC-10 headset-microphone shall provide an MTBF of 1500 hours during service life.

3.5.2 Longevity (Total Operating Life).- The H-159/GIC-10 headset-microphone shall have a longevity or minimum operating life of 5000 hours before wearout failures occur or the MTBF becomes less than the value specified.

3.6 Performance of Complete Headset-Microphone Assembly

3.6.1 D-C Resistance of Headset Circuit.- The d-c resistance of the headset circuit with the switch of Connector-Switch U-177/U in the center position shall be between 80 and 120 ohms. With the switch in either extreme position; the d-c resistance shall be between 21 and 46 ohms.

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3.6.2 D-C Resistance of Microphone Circuit.- The d-c resistance of the microphone circuit with the switch of the Connector-Switch U-177/U in the center position shall be between 530 and 570 ohms. With the switch in either extreme position, the d-c resistance shall be in excess of 1 megohm.

3.6.3 Talk-Out.- Each headset-microphone shall operate satsifactorily when subjected to the talk-out test specified herein.

3.6.4 Articulation Score.- The headset-microphone shall meet the articulation requirements of MIL-I-27185 when tested as specified herein.

3.7 Headband Headset.- The force exerted by the headband on the head of the user shall not be less than 4 pounds with maximum extension of the slide adjustment and not more than 7 pounds with minimum extension. The sliding action between the slide bearings of the earcup and the headband springs shall be smooth and uniform throughout the entire extension distance. The force required to move the earcup over the springs shall be between 3 and 6 pounds. After subjection to any or all of the environmental tests or the endurance test specified herein, this force shall not be less than 80 percent of the initial force.

3.8 Earcup, Sound Protector.- The sound attenuation properties of the earcup shall not differ appreciably from the mean curve shown in figure 1. The mean sound attenuation curve of a sample population (5 subjects minimum) shall not be less than the lower limits shown in figure 1 at any of the test frequencies.

3.9 Earphone, Sound Powered H-181/GIC-10.- The outline dimensions and exterior finish of the earphone shall be in accordance with figure 2.

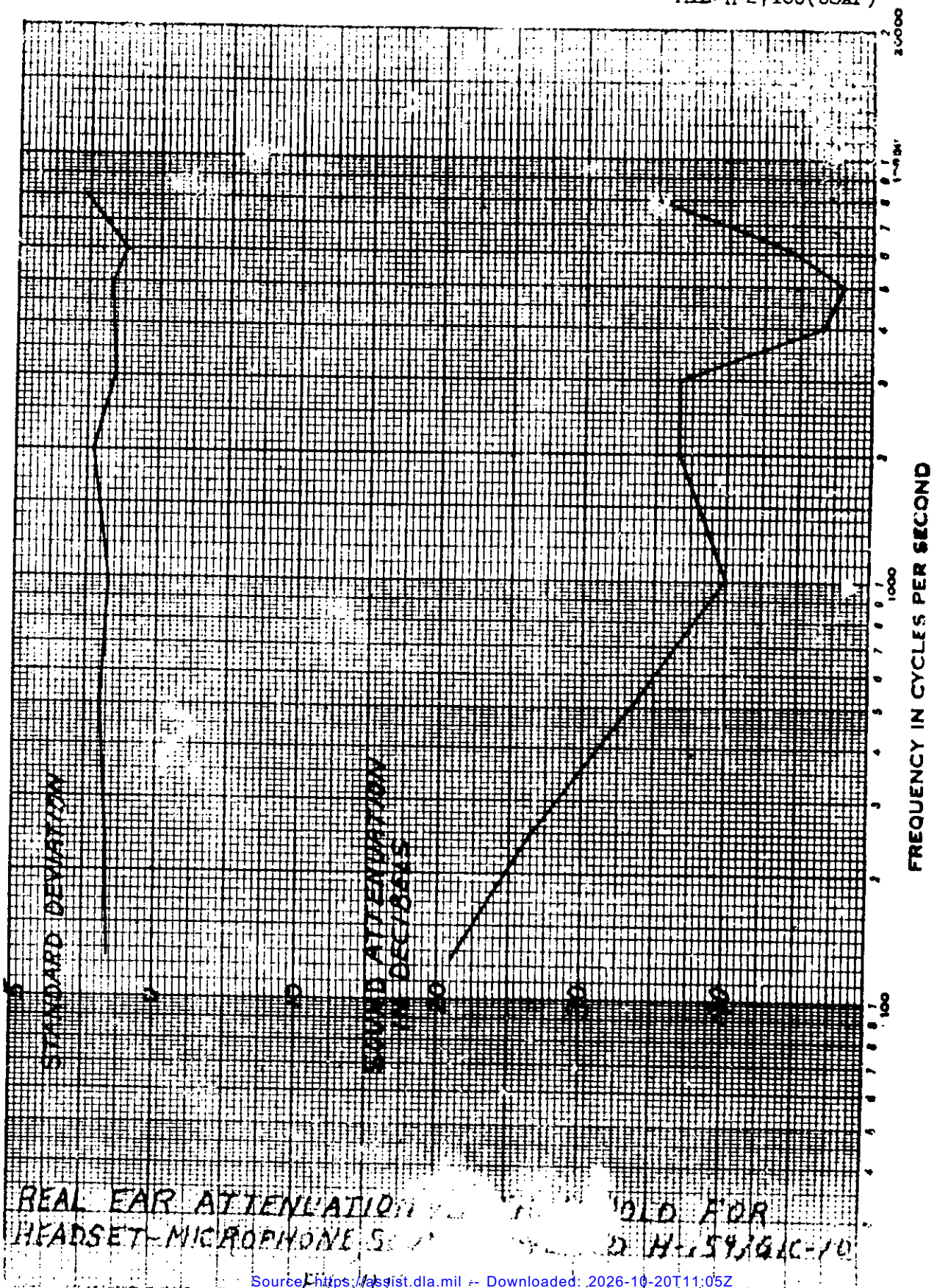
3.9.1 Impedance of Earphone.- The impedance of Earphone, Sound Powered H-181/GIC-10 at 1,000 cycles per second at 20°C shall be 1200 ohms plus or minus 10 percent.

3.9.2 Dielectric Strength of Earphone.- With the voice coil leads connected to the earphone terminals, the insulation between the case and each terminal shall withstand application of a 25- to 60-cps alternating-current potential having an rms value of 50 volts without flashover or breakdown.

3.9.3 Sensitivity of Earphone.- The sensitivity of the earphone at 1000 cps at ground level shall not be less than 118.0 decibels above a reference level of 0.0002 dyne per square centimeter.

3.9.4 Pressure Response-Frequency Characteristic of Earphone.- The pressure response-frequency characteristic between 100 and 4000 cps shall be within the limits of the response envelope shown on figure 3. The response in the 1000 to 3000 cps frequency range shall not change by more than 5.0 decibels in any 500 cycle increment.

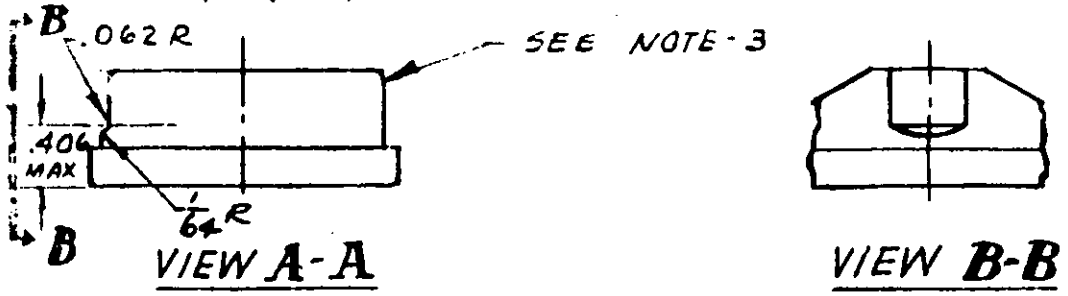
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REAL EAR ATTENUATION [illegible] HOLD FOR
HEADSET-MICROPHONE S[illegible] H-[illegible]

FIG. 2

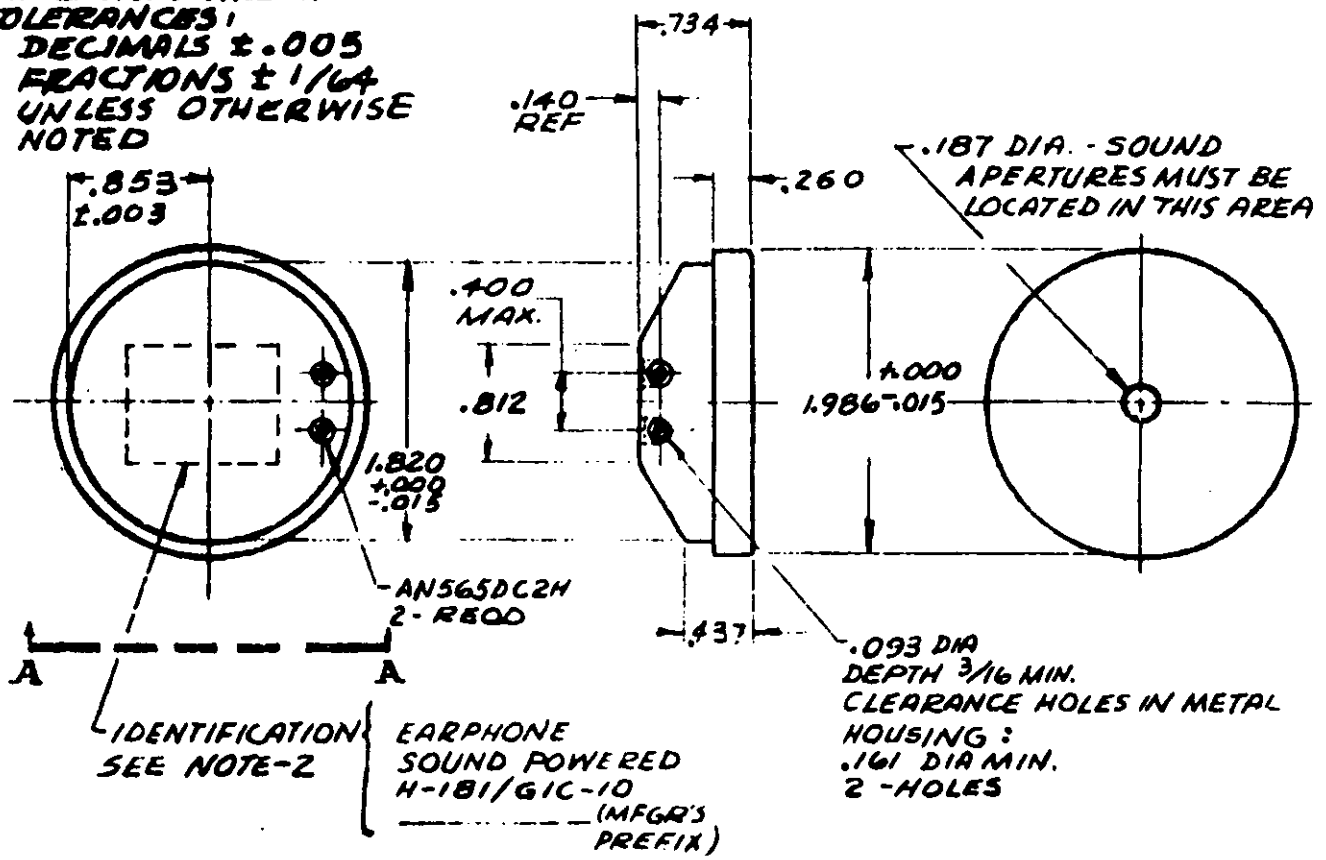
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3 ALL EXTERIOR SURFACES SHALL BE FINISHED PER USAF SPEC BUL F29-8, COLOR 1205

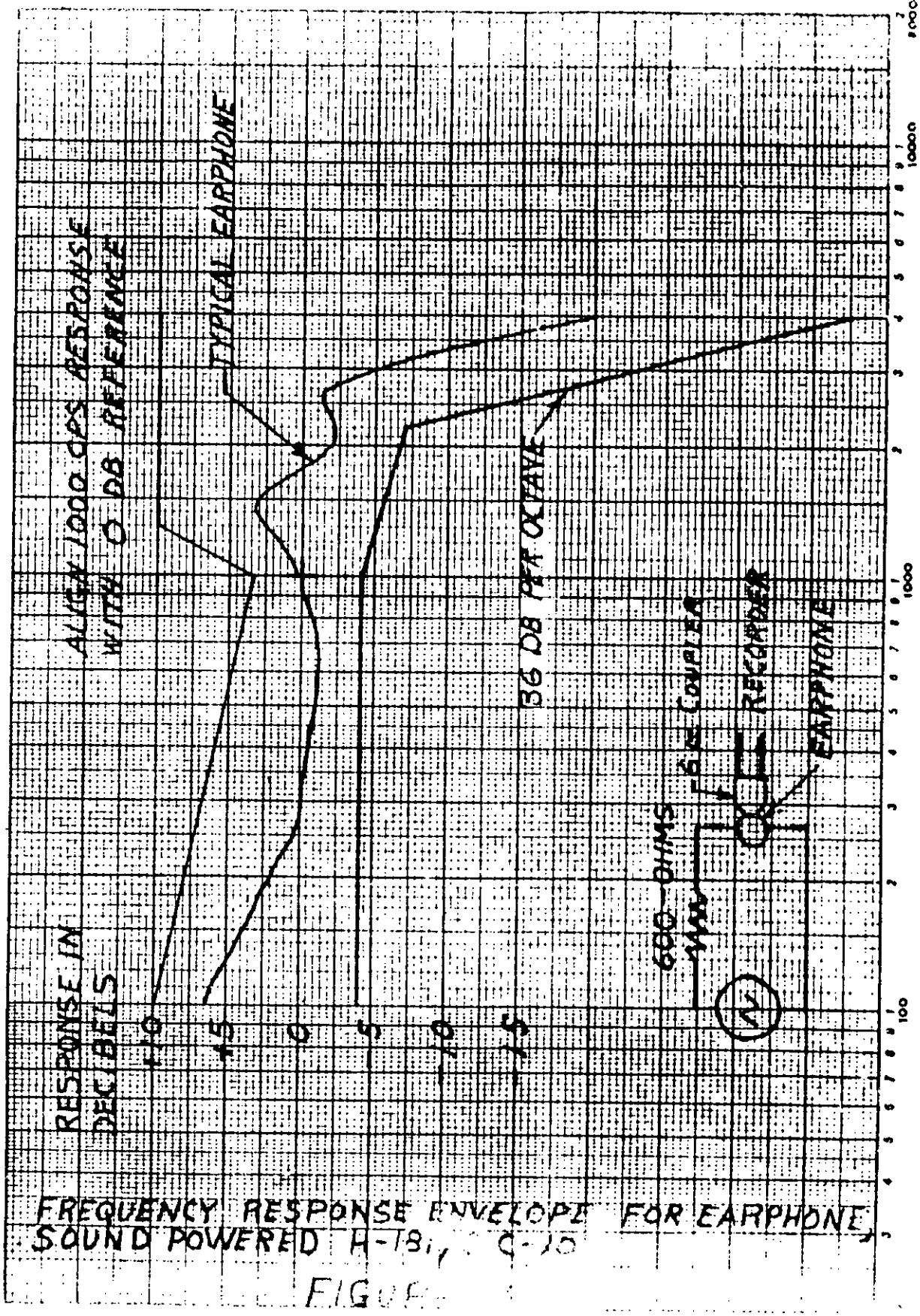
2 CHARACTERS SHALL BE 3/32 HIGH AND SHALL BE PERMANENTLY IMPRESSED

1 DIMENSIONS ARE IN INCHES.
TOLERANCES:
DECIMALS ±.005
FRACTIONS ±1/64
UNLESS OTHERWISE NOTED



EARPHONE, SOUND POWERED H-181/GIC-10
FIGURE 2

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FREQUENCY RESPONSE ENVELOPE FOR EARPHONE
SOUND POWERED H-18, C-55

FIGURE

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3.9.5 Pressure Response-Frequency Characteristic of Earphone in Earcup, Sound Protector.- The pressure response-frequency characteristic of the earphone mounted in an earcup shall not differ appreciably from that shown in figure 4 over the frequency range of 100 to 4000 cycles per second. The minimum sensitivity at 1000 cps shall not be less than 115 decibels above a reference level of 0.0002 dyne per square centimeter.

3.9.6 Overload Capability of Earphone.- The earphone shall be capable of withstanding a signal of random noise of approximately 20 milliwatts (5.0 volts) for a minimum of 24 hours.

3.9.7 Armature Stability of Earphone.- The armature of the earphone shall be capable of withstanding a mechanical compression against the pole piece without any degradation in performance below that specified herein. The impedance shall not vary more than plus or minus 100 ohms and the pressure response-frequency characteristic shall not vary more than plus or minus 3 decibels as a result of the armature stability test specified herein.

3.10 Microphone Sound Powered M-89/GIC-10.- The outline dimensions and exterior finish of the microphone shall be in accordance with figure 5.

3.10.1 Impedance of Microphone.- The impedance of the Microphone, Sound Powered M-89/GIC-10 at 1000 cycles per second at 20°C shall be 200 ohms plus or minus 10 percent when measured as specified herein.

3.10.2 Dielectric Strength of Microphone.- With the voice coil leads connected to the microphone terminals, the insulation between the case and each terminal shall withstand the application of a 25 to 60 cps alternating-current potential having an rms value of 50 volts without flashover or breakdown.

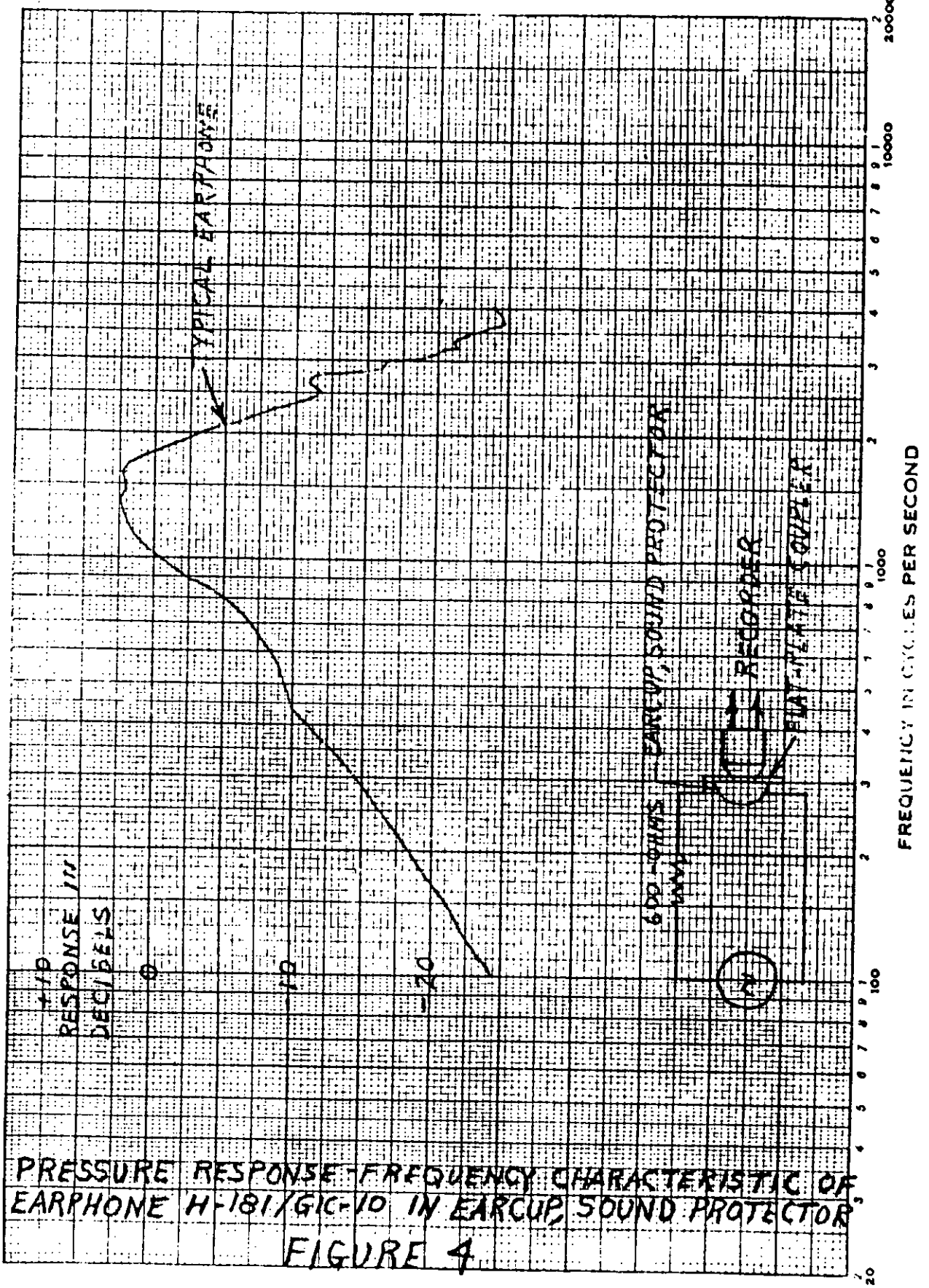
3.10.3 Sensitivity of Microphone.- The average open-current sensitivity of the microphone measured at ground level between 1000 and 2000 cps at 20°C (68°F) shall not be less than 0.25 millivolt.

3.10.4 Response-Frequency Characteristic of Microphone.- The response-frequency characteristic of the microphone at ground level, between 200 and 4000 cps, shall be within the response envelope shown in figure 6.

3.10.5 Directional Characteristic of Microphone.- The microphone shall show a difference of at least 15 decibels in output between the front response and the response at two angles of approximately 90 and 270 degrees with respect to the front.

3.10.6 Armature Stability of Microphone.- The armature of the microphone shall be capable of withstanding a mechanical compression against the pole piece without any degradation in performance below that specified herein. The impedance shall not vary more than plus or minus 20 ohms and the response frequency characteristic shall not vary more than plus or minus 3 decibels as a result of the armature stability test specified herein.

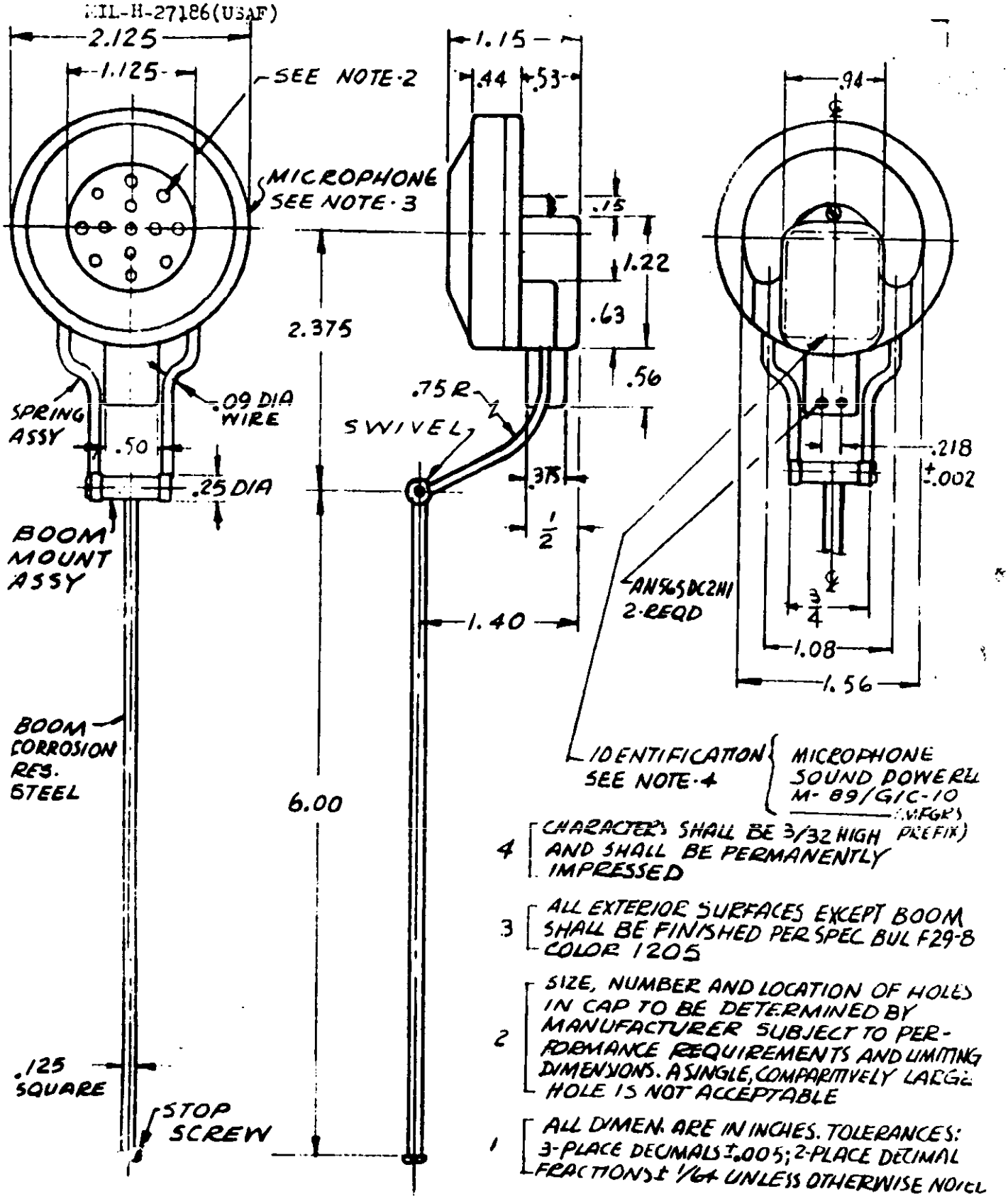
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PRESSURE RESPONSE-FREQUENCY CHARACTERISTIC OF EARPHONE H-181/GIC-10 IN EARCUP SOUND PROTECTOR

FIGURE 4

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4. CHARACTERS SHALL BE 3/32 HIGH AND SHALL BE PERMANENTLY IMPRESSED

3. ALL EXTERIOR SURFACES EXCEPT BOOM SHALL BE FINISHED PER SPEC BUL F29-8 COLOR 1205

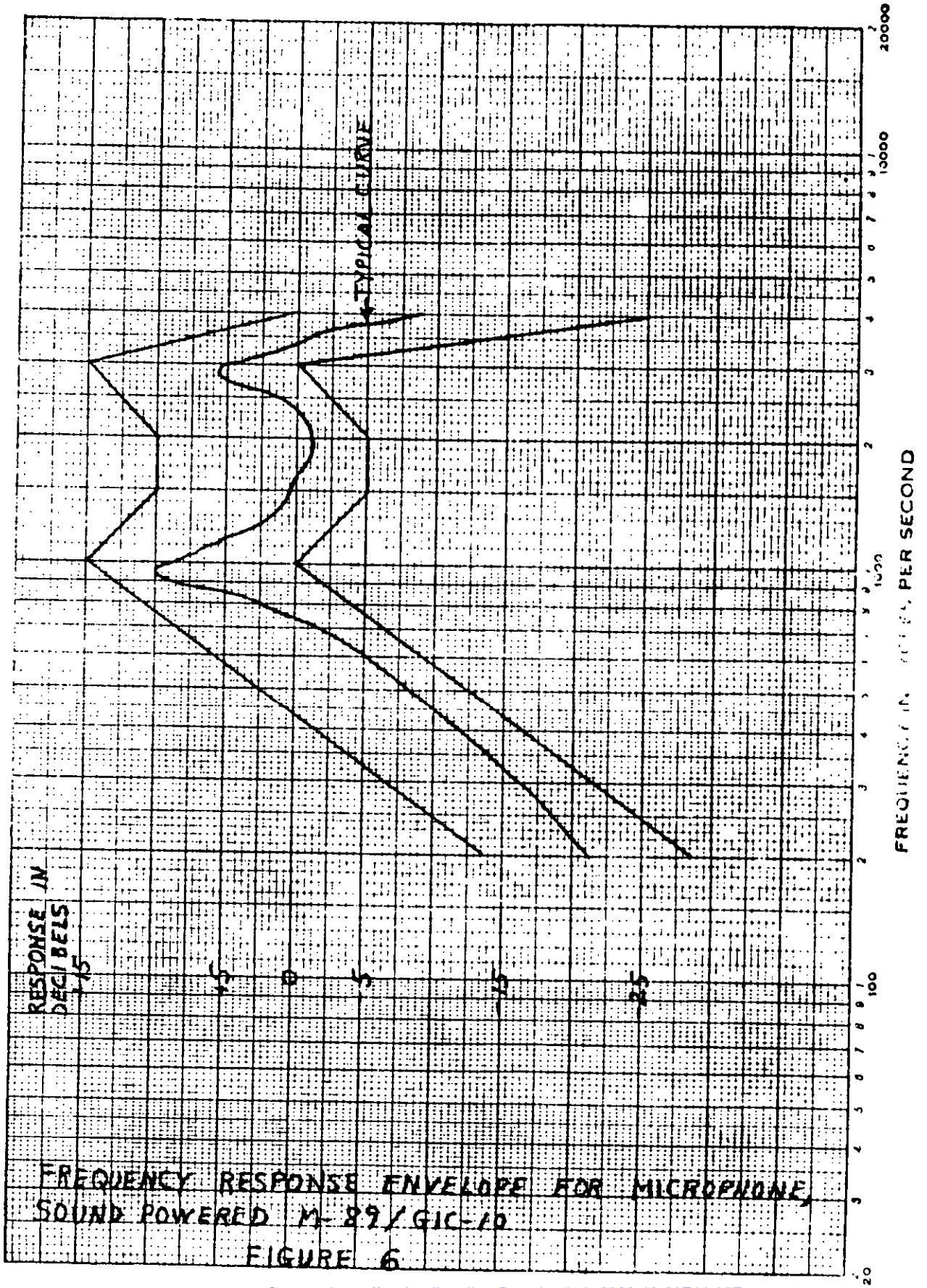
2. SIZE, NUMBER AND LOCATION OF HOLES IN CAP TO BE DETERMINED BY MANUFACTURER SUBJECT TO PERFORMANCE REQUIREMENTS AND LIMITING DIMENSIONS. A SINGLE, COMPARITIVELY LARGE HOLE IS NOT ACCEPTABLE

1. ALL DIMEN. ARE IN INCHES. TOLERANCES: 3-PLACE DECIMALS ±.005; 2-PLACE DECIMAL FRACTIONS ± 1/64 UNLESS OTHERWISE NOTED

MICROPHONE, SOUND POWERED M-89/GIC-10

FIGURE 5

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FREQUENCY RESPONSE ENVELOPE FOR MICROPHONE, SOUND POWERED M-89/GIC-10

FIGURE 6

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3.10.7 Microphone Bracket.- The sliding action between the microphone boom and the boom mounting bracket shall be smooth and uniform for the entire adjustment length. The force required to move the boom through the mounting bracket slide shall be between 2 and 5 pounds. The pivot action of the microphone bracket shall be smooth and uniform and be capable of rotating for 350 degrees. The force required to rotate the boom shall be between 40 and 60 ounce-inches. The slide force and rotation torque shall not be reduced by more than 20 percent when subjected to the endurance test specified herein.

3.11 Cord Assembly

3.11.1 Construction

3.11.1.1 Conductors.- The nominal area of the conductor shall be 320 circular mils and the minimum area 310 circular mils. Each conductor shall consist of bunch-stranded AWG No. 40 (0.003145 inch diameter) silver-plated cadmium bronze wire. The strands shall be twisted to have a righthand lay of approximately one-half inch. The individual strands, before bunching and plating, shall have a minimum tensile strength of 90,000 pounds per square inch and have a conductivity grade of at least 80 percent.

3.11.1.2 Insulation.- The insulation shall consist of 2 wraps of unsintered teflon tape well overlapped and applied in opposite directions. The finished conductor insulation shall be sintered to form the insulation into a homogeneous mass. The total composite insulation shall not be less than 0.006 inch thick. The insulation shall withstand a potential of 1000 volts of a commercial line frequency when applied in accordance with method 6211 of J-C-98.

3.11.1.3 Color Code.- For circuit identification, the insulation for the 6-conductor cord assembly shall be colored white, black, red, green, black, and white in that order.

3.11.1.4 Stay Cords.- The stay cords shall be composed of extra strength rayon threads braided together; they shall not abrade or otherwise damage the insulation of the conductors when the cord assembly is bent or stretched.

3.11.1.5 Jacket.- The cable shall be covered with a braided jacket composed of black nylon, 210 denier, 5 ends, 24 carrier, approximately 24 picks per inch.

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3.11.2 Physical Characteristics

3.11.2.1 Temperature Range.- The cable shall be flexible and resilient throughout the temperature range of minus 62°C to plus 85°C and shall not show evidence of cracking or other damage as a result of exposure to the temperature range specified herein.

3.11.2.2 Flexing Life.- The 6-conductor cable shall have a mean flex life of 50,000 flexes without showing evidence of:

a. Damage to the outside jacket.
b. Damage to the insulation of the individual conductors.
c. Electrical discontinuity.

3.11.3 Electrical Characteristics

3.11.3.1 Dielectric Strength.- The complete cord assembly shall meet the dielectric strength requirements of MIL-I-4997.

3.11.3.2 Insulation Resistance.- The complete cord assembly shall meet the insulation resistant requirements of MIL-I-4997.

3.11.3.3 Isolation.- The electromagnetic and electrostatic isolation between the headset circuit and the microphone circuit of the cable assembly shall not be less than that shown in table I.

TABLE I

Frequency Cycles per Second	Electrostatic Isolation Less Connectors	Electromagnetic With Connectors	Isolation Less Connectors
20,000	115 db	86 db	94 db
10,000	121 db	92 db	92 db
5,000	127 db	98 db	98 db

3.11.4 Conductor Strain Relief.- The cable shall be anchored at the headset and plug connectors in such a way that the cable can withstand the strain-relief test specified herein.

3.11.5 Connector-Switch U-177/U.- The design, construction, and performance of Connector-Switch U-177/U used on the cord assembly shall be in accordance with MIL-C-27187.

3.12 Color.- The plastic parts of the headset-microphone such as the earcup, headband pad, earcup pad, and noise shield shall be in accordance with USAF Specification Bulletin F29, color 1205 (International Orange).

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3.13 Weight.- The weight of the Headset-Microphone, Sound Powered H-159/GIC-10 shall not exceed 1 pound, 5 ounces.

3.14 Government-Loaned Property.- When provided in the contract or purchase order, and if available, the following items will be loaned by the Government to the contractor, upon his request, for use as a guide in designing the headset-microphone:

Item	Description	Quantity
1	Headset-Microphone, Sound Powered H-159/GIC-10	2
2	Cord Assembly, Electrical CX-4407/U	5
3	Interconnecting Box J-893/GIC-10	1
4	Interphone Control C-823A/AIC-10	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of the microphone shall be classified as follows:

a. Qualification tests..... See 4.5.
b. Acceptance tests..... See 4.6.

4.2 Test Conditions.- Unless otherwise specified, the microphone shall be tested under the following conditions, with moisture barrier and moisture barrier guard in place:

Temperature: Room ambient, plus 15°C to plus 35°C
Pressure: Normal atmospheric
Humidity: Prevailing ambient up to 90 percent relative humidity

4.2.1 Acoustical Tests on Earphone Sound Powered H-181/GIC-10.- Acoustical tests shall be made in general accordance with American Standards Association Standard Z24.9-1949, modified as necessary to obtain the acoustical characteristics in terms of those specified herein. All measurements shall be made using a source of constant available power. The source impedance shall be resistive and shall have a value of 1200 ohms over the frequency range of 100 cps to 10,000 cps.

4.2.1.1 Six-Cubic Centimeter Test Coupler.- The 6-cc coupler shall be in accordance with the type 1 coupler as specified in ASA Standard Z24.9-1949 for earphones which rest against the pinna. The dimensions which control the proper seating and sealing of the earphone and the measuring microphone to the coupler may be modified provided that the volume of the coupler is maintained at 6 cc.

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4.2.2 Acoustical Tests on Microphone, Sound-Powered M-89/GIC-10

4.2.2.1 Simulator Microphone Signal (Artificial Voice).- An artificial voice as shown in Drawing No. 58B12627 shall be used as the sound source in performing the sensitivity and close-response frequency-characteristic measurement specified herein. The artificial voice shall be calibrated with a Western Electric 640AA microphone, or demonstrated equivalent, which shall have been given standard calibration by the reciprocity technique.

4.2.2.2 Test Fixtures.- The fixtures used to position the calibrating microphone of the test microphone with respect to the sound source shall be designed to minimize any possible error in acoustical measurements owing to vibration, sound transmitted from the voice tube, unstable positioning of microphones, reflections from fixtures or environment (such as the interior of an altitude chamber), and ambient noise. No baffles shall be used near the sound-sensitive parts of the microphones; the test fixture shall be so designed that there is no possibility of error in acoustical measurements owing to sound reflection from surfaces of the test fixture.

4.2.2.3 Reference Sound Pressure Level.- All measurements shall be made at such a sound pressure level that the harmonic distortion or the noise produced in the test microphone by the air pumped through the orifice of the artificial voice at low frequencies shall be sufficiently small so as not to introduce a significant measurement error. For the purpose of this specification, all sensitivity data, regardless of the actual sound pressure level used in the measurements, shall be referred to a sound pressure level of 74 db relative to 0.0002 dyne per square centimeter.

4.2.2.4 Calibration of Close Sound Source.- The sound source shall be calibrated to produce a desired sound pressure at the grid of the 640AA microphone over the frequency range of 200 cps to 4000 cps using the pressure response calibration of the 640AA microphone. The grid of the 640AA microphone shall be spaced one-quarter inch from and coaxial with the mouth of the artificial voice during calibration.

4.2.2.5 Sensitivity and Response-Frequency Characteristic Measurements, Close Sound Source.- The test fixture employed to hold the microphone under test shall position the microphone in such a way that the sound port is 1/4 inch from and coaxial with the opening in the artificial voice. This 1/4 inch shall be measured to the microphone case itself.

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4.2.2.6 Sensitivity to a Close Sound Source.- The sensitivity of the microphone at ground level shall be determined by the minimum voltage developed and any frequency between 1000 cps and 2000 cps for a constant sound pressure level input. The output voltage developed by the microphone shall be measured on a Ballantine Electronic Voltmeter Model 300 or an equivalent having similar ballistic characteristics.

4.2.2.7 Response-Frequency Characteristic to a Close Sound Source.- The close-response-frequency characteristic of the microphone at ground level shall be obtained by measuring the open-circuit output voltage of the microphone over the frequency range of 200 cps to 4000 cps with a constant input sound pressure.

4.2.2.8 Plane Wave Tube.- A suitable plane wave tube shall be constructed having a means for supporting the microphone under test at least 24 inches from the sound source. The microphone shall be capable of rotation within the plane wave tube to determine its signal output at 90 degrees and 270 degrees with respect to the signal output when the front face of microphone is prependicular to the sound source.

4.2.3 Random Noise.- The random noise for testing earphones and microphones specified herein shall be produced from a source having substantially a Gaussian amplitude distribution. The spectrum (power per cycle) of the noise source shall be substantially uniform from 400 to 2000 cps and fall off above 2000 cps at a rate no less than 18 db per octave.

4.3 Test Methods

4.3.1 Qualification Reliability Test.- Seven samples shall be tested by the contractor for reliability and 2 of the 7 continued through a longevity test in accordance with the pilot production procedures of Task Group 3 of the AGREE report using test level H except that the vibration shall be at approximate constant frequency in the range of 20 to 60 cycles per second with a total excursion of 0.030 inch or an acceleration of plus or minus 2g, whichever is applicable. The motion need not be simple harmonic or linear. Duration of the MTBF demonstration test on each equipment shall not be less than 3 times the specified MTBF. Acceptance, permitting production, shall be judged on the basis of reliability test criteria (table entitled "Reliability Test Criteria") of AGREE Task Group 3. Recording, data handling, and reporting procedures shall be in accordance with the recommendations of AGREE Task Group 3.

4.3.2 Examination of Product.- Each headset-microphone shall be examined to determine compliance with the requirements of this specification for materials, workmanship, dimensions, and markings. Particular attention shall be paid to the following:

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a. Headband-earcup slide adjustment.
b. Elasticity of headband.
c. Swivel adjustment of microphone bracket.
d. Soldering operations.
e. Screw assemblage.
f. Color code of conductor insulation.

4.3.3 Headband, Headset.- The force exerted by the headband shall be measured at both the maximum and minimum extension. The distance shall be maintained at 8-1/4 inches between the plastic beads fastening the earcup to the headband. The force requirement to slide the earcup over the headband springs shall be measured for initial reference. The earcup shall be moved back and forth over the entire adjustable distance on the headband springs for a minimum of 10,000 times. After this test, the force required to slide the earcup over the spring shall be within the limits specified herein.

4.3.4 Earcup, Sound Protector.- The sound attenuation properties of the earcup shall be determined by the method outlined in ASA Standard Z24.22-1957 "Proposed American Standard Method for the Measurement of the Real-Ear Attenuation of Ear Protectors at Threshold." A minimum of 5 subjects shall be used in obtaining the mean sound attenuation curve.

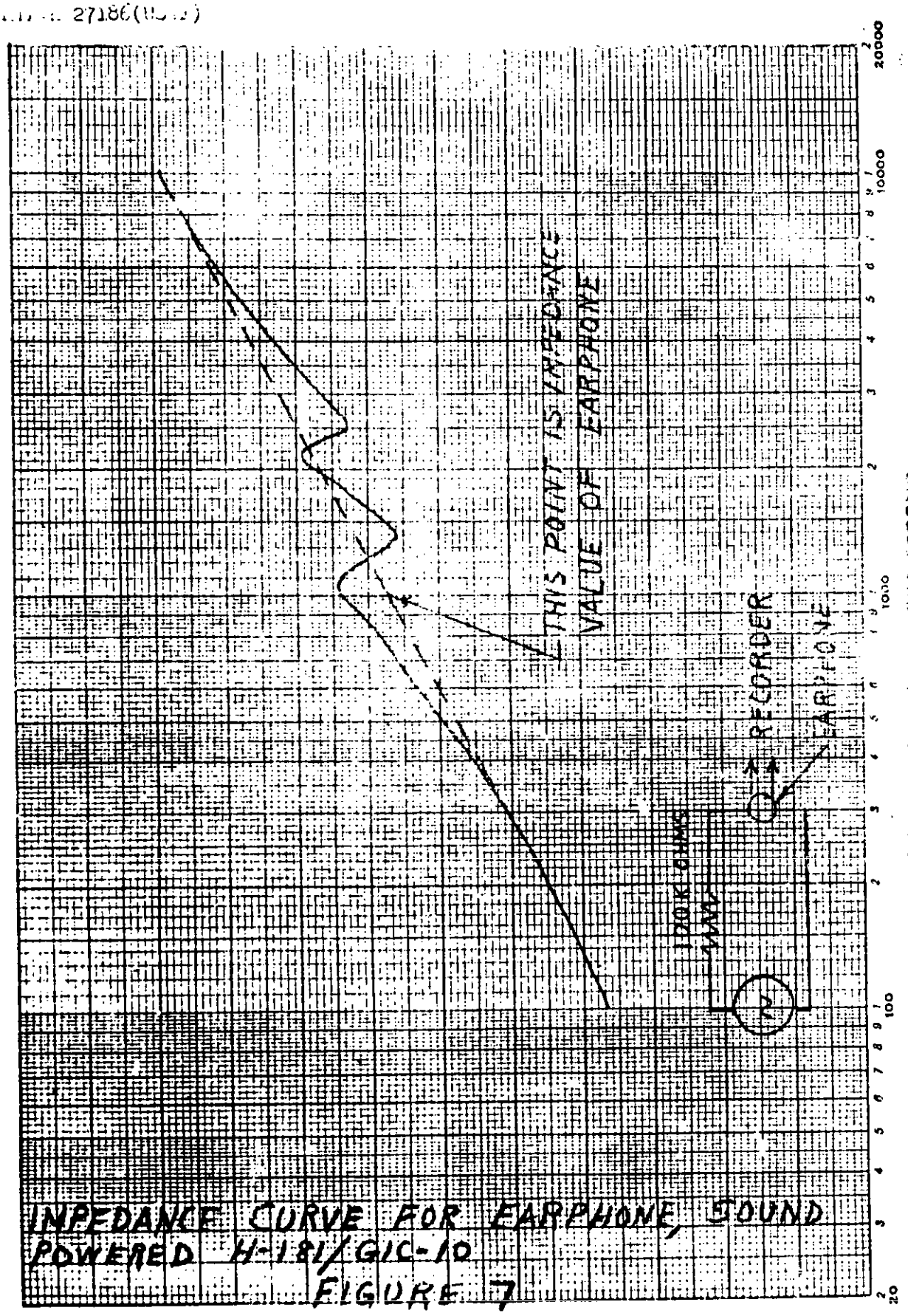
4.3.5 Earphone, Sound Powered H-181/GIC-10

4.3.5.1 Inpedance of Earphone.- The impedance of the earphone shall be measured with an audio oscillator driving the earphone with a 100,000 ohm resistor in series with the earphone. The voltage shall be set up so that 10 microamps of current flows through the circuit. A curve shall be drawn of the voltage across the earphone from 100 to 10,000 cps. A straight line shall be drawn tangent to the curve below and above the resonant frequency of the earphone (see figure 7). The point where this line crosses the 1000 cps line shall be measured and taken as the impedance value of the earphone.

4.3.5.2 Dielectric Strength of Earphone.- A 25-60 cps alternating potential having an rms value of 50 volts shall be applied between each terminal and a conductive area of the case for a minimum of 2 minutes.

4.3.5.3 Sensitivity of Earphone.- The sensitivity of the earphone shall be determined at 1000 cps with constant power of 1 millivolt.

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INPEDANCE CURVE FOR EARPHONE, SOUND POWERED H-181/GIC-10

FIGURE 7

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4.3.5.4 Pressure Response-Frequency Characteristics of Earphone at Ground Level.- The pressure response-frequency characteristic shall be measured as a continuous function of frequency over the range of 100 to 4,000 cps for a constant power available of 1 milliwatt. The 1000 cps point of the frequency response curve shall be aligned with the 1000 cps zero decibel reference point of figure 3 to determine whether the response curve is within the specified response envelope.

4.3.5.5 Pressure Response-Frequency Characteristic of Earphone in Earcup, Sound Protector.- The earphone shall be mounted in an Earcup, Sound Protector which shall be placed face down on the flat-plate coupler and centered over the microphone. The flat-plate coupler shall be a thick, rigid plate with a Western Electric Type 640AA microphone, or equivalent, mounted flush with the top surface. The pressure response frequency characteristic shall be measured as a continuous function of frequency over the range of 100 to 4,000 cps for a constant power available of 1 milliwatt.

4.3.5.6 Overload Capability of Earphone.- The earphone shall be operated continuously for 24 hours using the random noise specified herein. The true rms value of the noise applied to the earphone shall be 5.0 volts.

4.3.5.7 Armature Stability of Earphone.- The armature shall be forced downward mechanically until it touches the pole piece. This is accomplished by placing a blunt probe against the driverbar nut that is accessible through the hole in the front cover and forcing it downward approximately 0.010 inch. The impedance and pressure response frequency characteristic shall be measured before and immediately after this test.

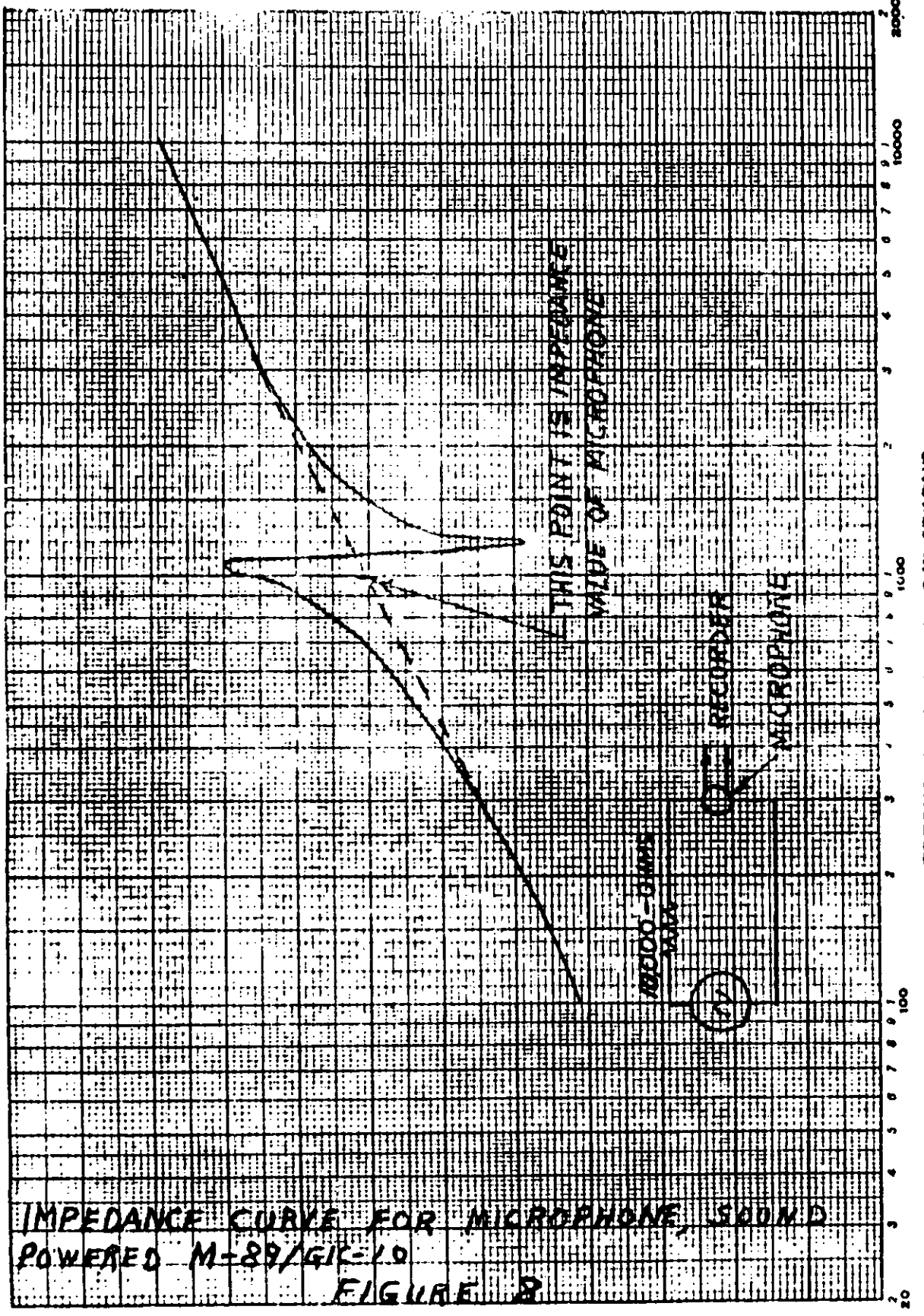
4.3.6 Microphone, Sound Powered M-84/GIC-10

4.3.6.1 Inpedance of Microphone.- The impedance of the microphone shall be measured with an audio oscillator driving the microphone with a 1,000 ohm series resistor. A voltage shall be set up so that 10 microamps of current flows through the circuit. A curve shall be drawn of the voltage developed across the microphone from 100 to 10,000 cps. A straight line shall be drawn tangent to the curve below and above the resonant frequency of the microphone (See figure 8). The point where this line crosses the 1000 cps line shall be measured and this shall be taken as the impedance value of the microphone.

4.3.6.2 Dielectric Strength of Microphone.- A 25-60 cps alternating potential having an rms value of 50 volts shall be applied between each terminal and a conductive area of the case for a minimum of 2 minutes.

4.3.6.3 Sensitivity of Microphone.- The sensitivity of the microphone at ground level shall be measured in accordance with the procedure specified herein.

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IMPEDANCE CURVE FOR MICROPHONE, 600 OHM POWERED M-89/GIC-10

FIGURE 2

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4.3.6.4 Response-Frequency Characteristic of Microphone.- The close-response frequency characteristic shall be measured at ground level in the form of a continuous curve in accordance with the test procedures specified herein. The response-frequency characteristic shall be checked against production limits which assure that the individual microphones meet the close-response frequency requirements specified herein.

4.3.6.5 Directional Characteristic of Microphone.- The directional characteristic of the microphone shall be measured using the random noise and plane-wave tube specified herein. The level of the signal at the sound source shall be adequate to provide enough output from the microphone to determine whether the difference is within the limits specified herein.

4.3.6.6 Armature Stability of Microphone.- The armature of the microphone shall be mechanically forced downward until it touches the pole piece. This is accomplished by placing a blunt probe against the driverbar nut that is accessible through the hole in the front cover and forcing it downward approximately 0.010 inch. The impedance and response-frequency characteristic shall be measured before and immediately after this test.

4.3.6.7 Microphone Bracket.- The force required to move the microphone boom through the boom mounting and the torque required to rotate the microphone boom in the microphone bracket shall be measured before and after the test specified herein. After the initial measurements have been recorded, the boom shall be made to slide its entire adjustment length through the boom mounting for a minimum of 25,000 times. The swivel adjustment shall be rotated through the entire 350 degrees for a minimum of 25,000 times.

4.3.7 Cord Assembly

4.3.7.1 Temperature Range.- The unterminated cable shall be placed within a test chamber wherein a temperature of plus 85°C is maintained for a period of 4 hours. While at this temperature, the cable shall be wrapped about a mandrel without damage to the cable jacket or conductor insulation. The diameter of the mandrel shall be the same as the nominal cable diameter. The cable shall be attached to the mandrel and suspended vertically with the lower end weighted sufficiently to keep the specimen taut and to permit wrapping without handling. The cable shall be inspected for damage or deformation and then subjected to a test chamber wherein a temperature of minus 62°C is maintained. After 20 hours of storage at minus 62°C, the cable shall be wrapped about the mandrel, while at this temperature, in the same manner as it was at the positive extreme temperature. The cable shall be examined for cracks or any other damage to the jacket or conductor insulation.

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4.3.7.2 Flexing Life.- Five samples of the 6-conductor unterminated cable shall be clamped and suspended through holes in a metal bar having a rectangular cross section of 1/2 inch by 5/8 inch as described below:

a. The clamping point shall be immediately above the metal bar; the cable clamp shall rest on top of the metal bar.

b. The cable shall be clamped in such a manner that it will not turn in the hole during the flexing test.

c. The length of the through holes in the metal bar shall be 5/8 inch. The diameter of the through holes shall be 0.005 to 0.010 inch greater than the diameter of the cable undergoing test.

d. The through holes in the metal bar shall incorporate a 1/8 inch radius at the end from which the cable emerges (bending point).

e. A knot shall be tied near the free end of the cable and a 1-pound weight shall be hung from the knot to maintain the cable in a vertical position.

f. The 6 conductors of each sample cable, a small low-current-drain 6-volt lamp, and a suitable power source shall be connected in series to indicate electrical continuity. The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical at a rate of approximately 85 cycles per minute. The arithmetical mean of the flexing cycles of the 5 samples of the cable before electrical discontinuity occurs shall be taken as the flexing life of the cable.

4.3.7.3 Dielectric Strength.- The complete cord assembly shall be tested for dielectric strength in accordance with MIL-I-4997.

4.3.7.4 Insulation Resistance.- The complete cord assembly shall be tested for insulation resistance in accordance with MIL-I-4997.

4.3.7.5 Isolation.- The cable assembly shall be tested for electromagnetic and electrostatic isolation to determine compliance with the limits specified herein.

4.3.7.6 Conductor Strain Relief

4.3.7.6.1 Pull Test at Cable Clamp.- A pull of 30 pounds shall be applied for a minimum of 2 minutes to the cable in the direction of the axis of the cable as it passes through the cable clamp mounted on the headset.

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4.3.7.6.2 Pull Test at Connector-Switch U-177/U.- A pull of 25 pounds shall be applied for a minimum of 2 minutes to the cable in the direction of the axis of the Connector-Switch U-177/U.

4.3.7.6.3 Pull Test at Earcup, Sound Protector.- A pull of 8 pounds shall be applied for a minimum of 2 minutes to the cable at the point where it emerges from the Earcup Sound Protector.

4.3.8 Connector-Switch U-177/U.- All tests performed on the Connector-Switch U-177/U shall be in accordance with MIL-C-27187.

4.3.9 Complete Headset-Microphone Assembly

4.3.9.1 D-C Resistance of Headset Circuit.- The d-c resistance of the headset circuit shall be measured across contacts number 2 and 4 of Connector-Switch U-177/U.

4.3.9.2 D-C Resistance of Microphone Circuit.- The d-c resistance of the microphone circuit shall be measured across contacts number 1 and 3 of Connector-Switch U-177/U

4.3.9.3 Talk-Out.- The headset-microphone shall be subjected to the talk-out test in accordance with MIL-I-27185 as a component of a complete set, assembled in accordance with Drawing No. 59D12685.

4.3.9.4 Articulation Score.- The headset-microphone shall be subjected to the articulation tests specified in MIL-I-27185.

4.3.10 Service Conditions (Environmental).- The environmental tests called out herein shall be conducted in accordance with MIL-T-5422 except that the headset-microphone shall not be operating during any of the environmental tests. A complete mechanical inspection and talk-out test shall be conducted after each environmental test.

4.4 Test Routine.- The headset-microphone shall be subjected to the tests listed in table II in the quantities shown. The tests need not be conducted in the order listed.

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TABLE II

Inspection or Test	Requirement	Test Procedure	Qualification	Individual	Sampling
Reliability	3.5	4.3.1	7		
Examination of product		4.3.2	all	all	all
Headband, Headset	3.7	4.3.3	all		all
Earcup, Sound Protector	3.8	4.3.4	all		all
Impedance of earphone	3.9.1	4.3.5.1	all		all
Dielectric strength of earphone	3.9.2	4.3.5.2	all	all	all
Sensitivity of earphone	3.9.3	4.3.5.3	all	all	all
Pressure response-frequency characteristic of earphone	3.9.4	4.3.5.4	all		all
Pressure response-frequency characteristic of earphone in Earcup, Sound Powered	3.9.5	4.3.5.5	all		
Overload capability of earphone	3.9.6	4.3.5.6	all		all
Armature stability of earphone	3.9.7	4.3.5.7	all		all
Impedance of microphone	3.10.1	4.3.6.1	all		all
Dielectric strength of microphone	3.10.2	4.3.6.2	all	all	all
Sensitivity of microphone	3.10.3	4.3.6.3	all	all	all

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TABLE II

Inspection or Test	Requirement	Test Procedures	Qualification	Individual	Sampling
Response frequency characteristic of microphone	3.10.4	4.3.6.4	all		all
Directional characteristic of microphone	3.10.5	4.3.6.5	all	all	all
Armature stability of microphone	3.10.6	4.3.6.6	all		all
Microphone bracket	3.10.7	4.3.6.7	all		all
Temperature range	3.11.2.1	4.3.7.1	5 (unterminated cable only)		
Flexing life	3.11.2.2	4.3.7.2	5 (unterminated cable only)		
Dielectric strength	3.11.3.1	4.3.7.3	all		all
Insulation resistance	3.11.3.2	4.3.7.4	all		all
Isolation	3.11.3.3	4.3.7.5	5		
Conductor strain relief	3.11.4	4.3.7.6	all		all
D-c resistance of headset circuit	3.6.1	4.3.9.1	all	all	all
D-c resistance of microphone circuit	3.6.2	4.3.9.2	all	all	all
Talk-out	3.6.3	4.3.9.3	all	all	all
Articulation score	3.6.4	4.3.9.4	5		

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TABLE II

Inspection or Test	Requirement	Test Procedures	Qualification	Individual	Sampling
Service conditions, environmental		4.3.10			
Temperature and altitude			3		all
Humidity			3		all
Vibration			3		
Shock			3		
Salt spray			1		

4.5 Qualification Tests

4.5.1 Samples for Testing.- A minimum of 10 headset-microphones shall be subjected to the qualification tests in the quantities indicated in table II. At least 3 headset-microphones shall be subjected to all of the qualification tests except the salt spray test. More than one qualification test may be performed on any one headset-microphone provided it has not been damaged by a previous test.

4.5.1.1 Samples for the Procuring Activity.- In addition, 3 headset-microphones shall be submitted to the procuring activity along with the test report and with a pressure response-frequency characteristic at ground level for each of the earphones and microphones.

4.5.2 Qualification Test Program.- Prior to any formal qualification testing of the sample headset-microphones, the manufacturer shall submit a draft of his proposed test program to the procuring activity for approval. The draft shall include:

a. A list of all tests to be performed and complete procedures for each test including block or schematic diagrams.

b. A list of test equipment to be used, identified by manufacturer and type number.

c. Copies of the data record forms to be used in recording the test data.

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4.5.2.1 Qualification Reliability Test Program.- Within 90 days after receipt of contract and 30 days prior to initiation of testing or at such later date as the contracting officer may authorize, the contractor shall submit a detailed test program to the procuring activity for approval. Testing shall begin only after formal approval of this program by the procuring activity.

4.5.3 Qualification Test Report.- Upon completion of the qualification tests, three copies of a complete test report, prepared in accordance with MIL-T-9107, and including the name and grade of all materials used in the headset-microphone shall be furnished the procuring activity.

4.6 Acceptance Tests.- Acceptance tests shall consist of:

a. Individual tests.
b. Reliability test.
c. Sampling tests.

4.6.1 Individual Tests.- Each headset-microphone shall be subjected to the individual tests listed in table II.

4.6.2 Acceptance Reliability Test.- Items of equipment shall be tested by the contractor for reliability and longevity and acceptance judged in accordance with the production test procedures of AGREE Task Group 3 using test level H except that the vibration shall be at approximate constant frequency in the range of 20 to 60 cycles per second with a total excursion of 0.030 inch or an acceleration of plus or minus 2g, whichever is applicable. The motion need not be simple harmonic or linear.

4.6.2.1 Acceptance Reliability Test Program.- Within 90 days after receipt of contract and 30 days prior to initiation of testing or at such later date as the contracting officer may authorize, the contractor shall submit a detailed test program to the procuring activity for approval. Testing shall begin only after formal approval of this program by the procuring activity.

4.6.3 Sampling Tests.- Unless otherwise specified, sampling tests as indicated in table II shall be performed on one percent of the headset-microphones on order. The meadset-microphones shall be selected at random from quantities of headset-microphones which have passed the individual tests specified herein.

4.6.3.1 Rejection and Retest.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

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4.6.3.2 Individual Tests May Continue.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of the specification.

5. PREPARATION FOR DELIVERY

5.1 General.- Preparation for delivery shall be in accordance with MIL-P-17555, method IIb.

5.2 Shipment Marking.- Shipment marking nomenclature shall be Headset-Microphone, Sound Powered H-159/GIC-10.

6. NOTES

6.1 Intended Use.- Headset-Microphone, Sound Powered H-159/GIC-10 is intended for use as part of the Intercommunication Set AN/GIC-10 which is a ground interphone system for use by maintenance personnel while working on aircraft. The AN/GIC-10 system may be used with or without external power.

6.2 Ordering Data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. List of interchangeable components and assemblies.
c. Samples will be available.
d. Selection of applicable levels of preservation and packaging, and packing.

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that data.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from Communication and Navigation Laboratory, WCLNK, Wright-Patterson Air Force Base, Ohio.

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6.4 Engineering Data.- Headset-Microphone, Sound Powered H-159/GIC-10 supersedes the headset-microphone portion of the Telephone, Sound Powered Head-Chest Set Type SA (NAVY). The H-159/GIC-10 headset-microphone is functionally but not electrically or mechanically interchangeable with the SA Type Head-Chest Set.

6.5 Definitions

6.5.1 Failure.- A failure is a condition whereby any required operating capability of the equipment becomes degraded beyond that specified contractually.

6.5.2 Reliability.- Reliability is the probability of performing without failure a specified function under given conditions for a specified period of time.

6.5.3 Duration.- All references to "test time" are to be understood as meaning equipment "on" time, or equipment operating time for purposes of MTBF or longevity determination.

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